DT081A - Signal and Image Processing
Exercise 3 Solutions

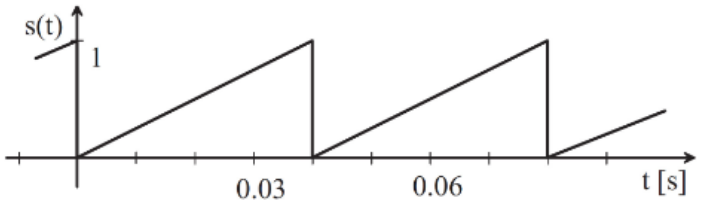
Question 1. A periodic signal can be described with a Fourier series.

a) Write the definition of the Fourier series and how the complex Fourier coefficients are calculated.

b) What does the Fourier series say about the composition of a periodic signal?

c) Consider the signal $s(t)$ in the figure below. What period T_0, basic frequency f_0, and angular frequency ω_0 does the signal $s(t)$ have?

d) Compute the Fourier components of the signal given below.



Answer:

a) The Fourier series is defined as: $s(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$, where $c_k = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} s(t)e^{-jk\omega_0 t}dt$.

b) The Fourier series shows that any periodic signal can be represented as a sum of sinusoids with frequencies that are integer multiples of the fundamental frequency.

c) REading the figure, we can see that $T_0 = 0.04$, meaning we have frequency $f_0 = 25$ Hz and $\omega_0 = 50\pi \text{s}^{-1}$.

d) The Fourier components of a sawtooth wave with period $T_0 = 0.04s$ and amplitude 1 are

$$c_0 = \frac{1}{T_0}\int_0^{T_0} \frac{t}{T_0}dt \tag{1}$$

$$= \frac{1}{2}. \tag{2}$$

For $k \in \mathbb{Z} \setminus \{0\}$, we get

$$c_k = \frac{1}{T_0}\int_0^{T_0} s(t)e^{-jk\omega_0 t}dt \tag{3}$$

$$= \frac{1}{T_0}\int_0^{T_0} \left(\frac{t}{T_0} - \left\lfloor \frac{t}{T_0} \right\rfloor\right) e^{-jk\omega_0 t}dt \tag{4}$$

$$= \frac{1}{T_0^2}\int_0^{T_0} te^{-jk\omega_0 t}dt \tag{5}$$

$$= \frac{1}{T_0^2}\left[-\frac{e^{-jk\omega_0 t}(jk\omega_0 t + 1)}{(jk\omega_0)^2}\right]_0^{T_0} \tag{6}$$

$$= \frac{1}{T_0^2}\left(\left(\frac{e^{-jk2\pi}(jk2\pi + 1)}{(k\omega_0)^2} - \frac{1}{(k\omega_0)^2}\right)\right) \tag{7}$$

$$= \frac{jk2\pi + 1 - 1}{(k2\pi)^2} \tag{8}$$

$$= j\frac{1}{k2\pi} \tag{9}$$

Question 2. The signal $s(t)$ (above) passes through an ideal low-pass filter with cut-off frequency $f_c = 60$ Hz. What frequencies are there in the output signal $y(t)$?

Answer: Noticing that s(t) has a frequency of 25 Hz, that means the output signal will only include frequency components of less than 60 Hz, meaning only we are left with the Fourier components c_k with $|k| < 60/25$.

Question 3. Decide the amplitude spectrum for the continuous-time signal $x(t)$ in the figure below.

Question 4. Decide the Fourier transform of the signal $x(t) = e^{-at}\sin(\omega_0 t)u(t)$. You can use a table.

Answer: Using Euler's identity, we get

$$x(t) = \frac{1}{2j}\left(e^{-(a-j\omega_0)t}u(t) - e^{-(a+j\omega_0)t}u(t)\right), \tag{10}$$

we can transform the two expressions separately and recombine them as follows

$$X(f) = \frac{1}{2j}\left(\frac{1}{a - j\omega_0 + j2\pi f} - \frac{1}{a + j\omega_0 + j2\pi f}\right) \tag{11}$$

$$= \frac{1}{2j}\left(\frac{1}{(a + j2\pi f) - j\omega_0} - \frac{1}{(a + j2\pi f) + j\omega_0}\right) \tag{12}$$

$$= \frac{1}{2j}\frac{2j\omega_0}{(a + j2\pi f)^2 + \omega_0^2} \tag{13}$$

$$= \frac{\omega_0}{(a + j2\pi f)^2 + \omega_0^2} \tag{14}$$

Question 5. Assume the system H to be linear.

a) What defines a linear system?

b) What defines a time-invariant system?

c) What defines a causal system? What is specific with its impulse response? Give an example of a difference equation of a non-causal discrete-time system.

Answer: For this exercise, see lecture 5 (slides 4 ++)

a) We call a system H linear, if it satisfies $H\{ax_1(t) + bx_2(t)\} = aH\{x_1(t)\} + bH\{x_2(t)\}$. This is also called the superposition principle.

b) A time-invariant system's response is independent of when the input is applied: if $y(n) = H\{x(n)\}$, then $y(n - n_0) = H\{x(n - n_0)\}$.

c) A causal system's output depends only on past and present inputs. Its impulse response $h(n) = 0$ for $n < 0$.

Question 6. The input to a LTI system contains the frequencies 5 Hz, 125 Hz and 1000 Hz. What are the frequency components of the output? Could there be fewer frequencies in the output?

Answer: The output will contain at most the same frequencies: 5 Hz, 125 Hz, and 1000 Hz, possibly with different amplitudes and phases. There could be fewer frequencies in the output if the system's frequency response is zero at any of these frequencies.

Question 7.

a) Write the convolution sum for a general discrete-time system.

b) Write the convolution sum for a causal discrete-time system.

c) Write the convolution sum for a general discrete-time system with a causal input signal.

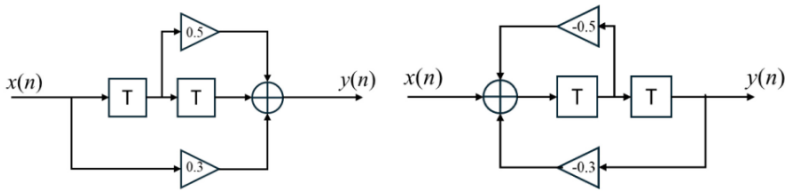
Answer:

a) General: $y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$

b) Causal: $y[n] = \sum_{k=0}^{\infty} h[k]x[n-k]$

c) General system, causal input: $y[n] = \sum_{k=-\infty}^{n} h[k]x[n-k]$

Question 8. Consider the discrete-time systems below.



a) Write the difference equations.

b) Compute the impulse response by iterating for $n = 0, 1, 2, 3, 4$

c) What is such a system called (considering the length of the impulse response).

Answer:

a) The difference equations are

$$y_1(n) = 0.3x(n) + 0.5x(n-1) + x(n-2) \tag{15}$$
$$y_2(n) = -0.3y(n-2) - 0.5y(n-1) + x(n-2) \tag{16}$$

b) for y_1, we get

$$\begin{aligned} 0 &\mapsto 0.3 \\ 1 &\mapsto 0.5 \\ 2 &\mapsto 1 \\ 3 &\mapsto 0 \\ 4 &\mapsto 0, \end{aligned}$$

while for y_2, we get

$$\begin{aligned} 0 &\mapsto 0 \\ 1 &\mapsto 0 \\ 2 &\mapsto 1 \\ 3 &\mapsto -0.5 \\ 4 &\mapsto -0.05 \end{aligned}$$

c) Both of these systems are called finite impulse response systems, as there exists an $N \in \mathbb{N}$, such that $y_1(n) = y_2(n) = 0$ for all n larger than N.

Question 9. A linear time-invariant, and causal discrete-time system is defined by its impulse response $h(n) = e^{0.5n}\sin(0.146n)u(n)$. Compute the output $y(n)$ for $n = 0, 1, 2, 3, 4$ by convolving the impulse response with the input signal $x(n) = \{x(0), x(1), x(2), x(3)\} = \{1, 2, 1, -1\}$

Answer: We compute

$$y(n) = \sum_{k=0}^{n} h(k)x(n-k) \tag{17}$$
$$= \sum_{k=0}^{n} e^{0.5k}\sin(0.146k)u(k)x(n-k) \tag{18}$$

and get

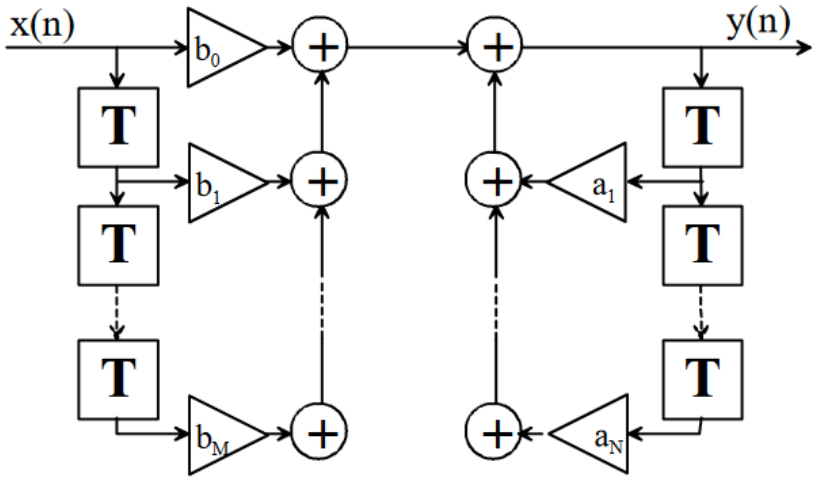
$$y(0) = 0 \tag{19}$$
$$y(1) = e^{0.5}\sin(0.146)\cdot 2 + 0 \approx 0.240 \tag{20}$$
$$y(2) = \cdots \approx 1.262 \tag{21}$$
$$y(3) = \cdots \approx 3.706 \tag{22}$$
$$y(4) = \ldots \approx 8.418 \tag{23}$$

Question 10. Draw the circuit diagram for a general IIR filter with M forward connections and N feedback connections.

a) Set the requirements on the parameters to assure it to be an IIR filter.

b) Set the requirements on the parameters to assure it to be an FIR filter.

Answer: The following diagram is depicted in lecture 5 slide 13



a) For an IIR filter, at least one feedback coefficient (a_k) must be non-zero and the forward and feedback connections must not cancel eachother out.

b) For an FIR filter, all feedback coefficients (a_k) must be zero (or they must be cancelled out), and at least one forward coefficient (b_k) must be non-zero.

Question 11.

a) What is the consequence of having a limited number of samples when carrying out a Fourier analysis?

b) Describe mathematically in both time domain and frequency domain the consequence of using $N < \infty$ samples of the signal $s(n)$, $0 \leq n < \infty$. What happens in both time and frequency domains when N is increased?

Answer:

a) Limited sampling can lead to leakage and reduced frequency resolution.

b) Time domain: $s_N(n) = s(n)w(n)$, where $w(n)$ is a rectangular window of length N. Frequency domain: $S_N(\omega) = S(\omega) * W(\omega)$, where $*$ denotes convolution. As N increases, the main lobe of $W(\omega)$ narrows, improving frequency resolution, meaning for infinite N, this would be a dirac (i.e. the identity w.r.t. convolution).

Question 12. Search for four different windows (for windowing) on the Internet. Give their respective mathematical expressions. Describe the size of main lob and sidebands for these.

a) Which of these windows would you choose to identify two neighbouring frequencies of about the same amplitudes?

b) Which of these windows would you choose to identify two neighbouring frequencies with rather different amplitudes?

Answer: Four common windows (found on Wikipedia: Window-function):

1. Rectangular:
$$w(n) = \begin{cases} 1 & \text{if } n \leq N, \\ 0 & \text{otherwise.} \end{cases}$$
The rectangular window has a narrow main lobe, offering high frequency resolution, but also high side lobes, leading to significant spectral leakage. This makes it challenging to distinguish closely spaced frequencies, particularly when their amplitudes differ.

2. Triangular (Bartlett):
$$w(n) = \begin{cases} 1 - \left|\frac{2n}{N} - 1\right| & \text{if } n \leq N, \\ 0 & \text{otherwise.} \end{cases}$$
The triangular (Bartlett) window has a wider main lobe than the rectangular window, which reduces frequency resolution but lowers side lobes. This provides better suppression of spectral leakage, making it easier to distinguish closely spaced frequencies of differing amplitudes, though at the cost of some resolution.

3. Sine:
$$w(t) = \begin{cases} sin(\frac{\pi n}{N}) & \text{if } n \leq N, \\ 0 & \text{otherwise.} \end{cases}$$
The sine window has a main lobe wider than both the rectangular and triangular windows, reducing frequency resolution, but its side lobes are significantly lower. This strong side lobe suppression minimizes spectral leakage, making it effective for distinguishing neighboring frequencies with different amplitudes, though at the expense of resolving closely spaced frequencies.

4. Welch:
$$w(n) = \begin{cases} 1 - \left|\frac{2n}{N} - 1\right|^2 & \text{if } n \leq N, \\ 0 & \text{otherwise.} \end{cases}$$

The Welch window has a wider main lobe compared to the rectangular and triangular windows, reducing frequency resolution, but it significantly lowers side lobes. This provides good suppression of spectral leakage, making it suitable for identifying neighboring frequencies with similar or differing amplitudes, though with some loss in the ability to resolve closely spaced frequencies.

a) For similar amplitudes, the rectangular window can be a good choice, as similar amplitudes reduce the risk of spectral leakage, meaning one may be able to better benefit of its narrow main lobe, but in practice a window with lower side-lobes may be preferred to decrease issues from spectral leakage (e.g. triangular window).

b) For different amplitudes the sine, strong side-lobe suppression is crucial, meaning the Welch or Sine window is to be preferred.

Question 13.

a) Derive the requirement on the impulse response $h(n)$ for a discrete-time system to be BIBO stable.

b) Let the impulse response of a discrete-time system be $h(n) = a^n$. Set the restriction on a for the system to be BIBO stable.

Answer:

a) For BIBO stability we require $\sum_{k=-\infty}^{\infty} |h(k)| < \infty$ (lecture 5 slide 23).

b) For $h(n) = a^n$, the system is BIBO stable if $|a| < 1$.

Question 14. A signal is sampled in four points, and one gets the sample values 1, -1, 0 and 1. Compute the DFT.

Answer: Using the formula (Lecture 1 slide 50)

$$X(k) = \sum_{n=0}^{3} x(n) e^{-j\frac{2\pi kn}{4}}, \tag{24}$$

we get

$$X(0) = 1 \tag{25}$$
$$X(1) = 1 + 2i \tag{26}$$
$$X(2) = 1 \tag{27}$$
$$X(3) = 1 - 2i. \tag{28}$$